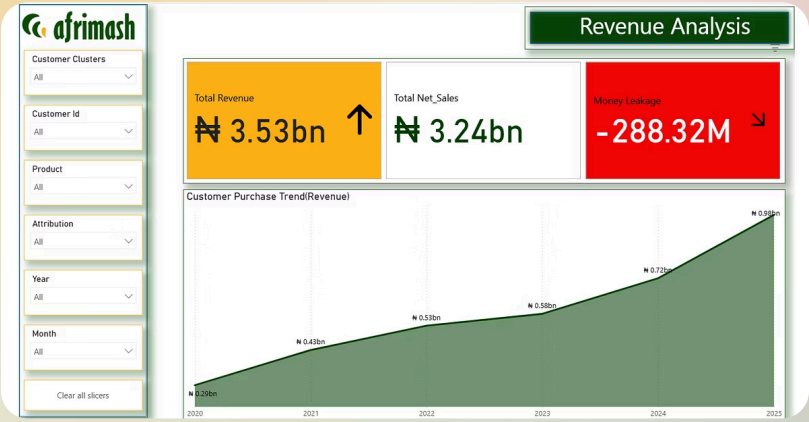


Afrimash Customer Intelligence — Process & Methodology

A comprehensive solution

Team RootCause

Made with GAMMA

Phase 1 — Scope

The Problem

- Challenges segmenting customers
- Lack of clear insights
- Generic product recommendations

Made with GAMMA

Data Sourcing & Preparation

	D
	Product(s)
2:50:44	1× Garri Processing Plant (Combo) - Locally Assembled | Free Shipping
4:52:40	8× AGRITED Pullets (Hyline Brown Pullets) Day-Old
2:46:29	1× AMO Broiler Chicks | Day-Old Birds
3:50:03	1× Papaya Nouna F1
4:36:11	5× Maradona F1 (Dwarf Papaya | 100 Seeds)
0:59:38	1× DK920 YELLOW Maize Seed – High-Yield | 30KG Bag | DK920 Per KG |DK777 WHITE
1:05:09	2× Iodine (Centre Brand | 10% Disinfectant) - 1L Keg
0:10:21	10× Seed Co Maize Seeds | (SC 719 Late Maturity) | (SC 649 Medium Maturing) | (SC5
6:05:02	
3:38:48	2× Commercial Day Old Pullets - ISA Brown (Grinphield Brand | Pre-Order)
6:19:38	1× Agrited Broiler | Commercial Day Old Broilers Ross 308
4:06:11	5× AGRITED Pullets (Hyline Brown Pullets) Day-Old
1:39:21	1× Vertex Broiler | Commercial Day-Old | Arbor acres plus
9:48:18	6× Agrited Pullet | Commercial Day Old Pullet - Hyline Brown (Pre-Order)
56:43	1× Vertex Broiler | Commercial Day-Old | Arbor acres plus

Sources

Transaction and RFM Data.

Afrimash

Data Cleaning

Exploded product column,

Removed Nulls for recommendation modeling

Base Feature Engineering

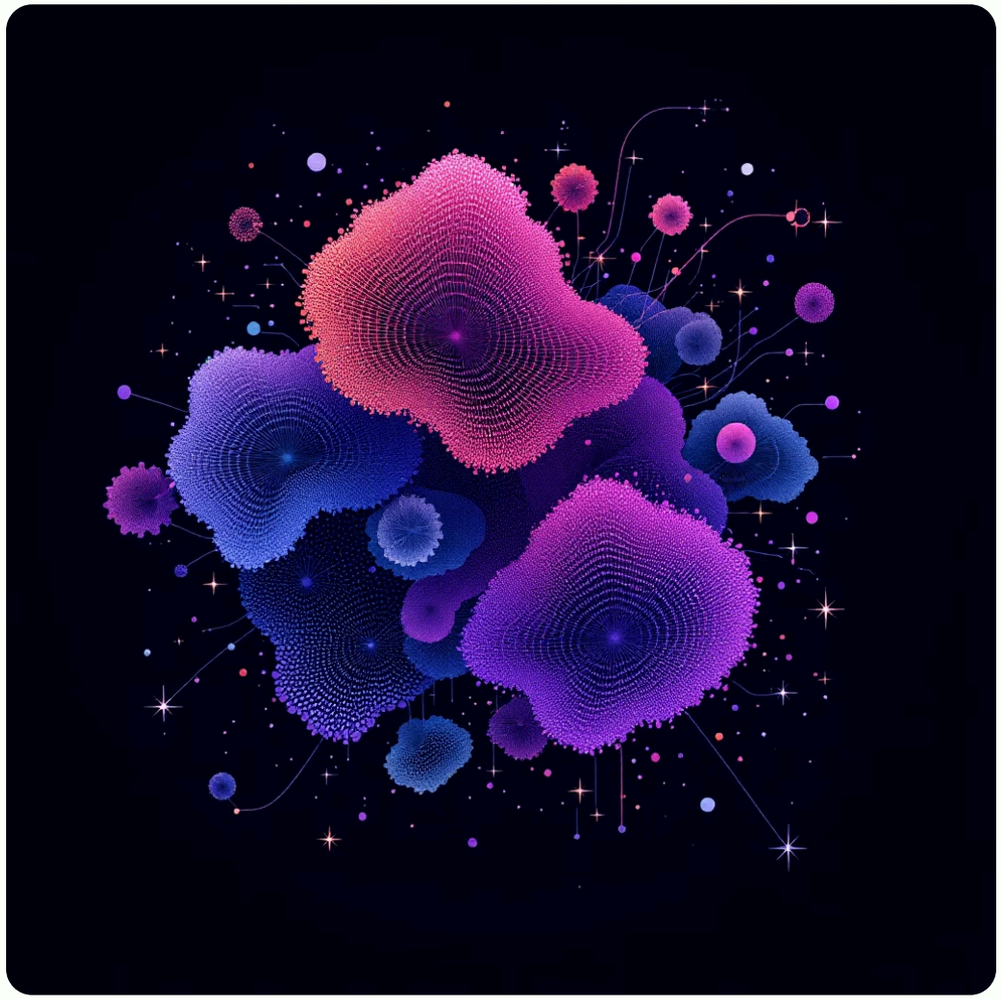
Created DAX measures, Repurposed the transaction data

Made with GAMMA

Phase 2 — RFM Behavioral Segmentation

Recency, Frequency and Monetary Metrics

K Means



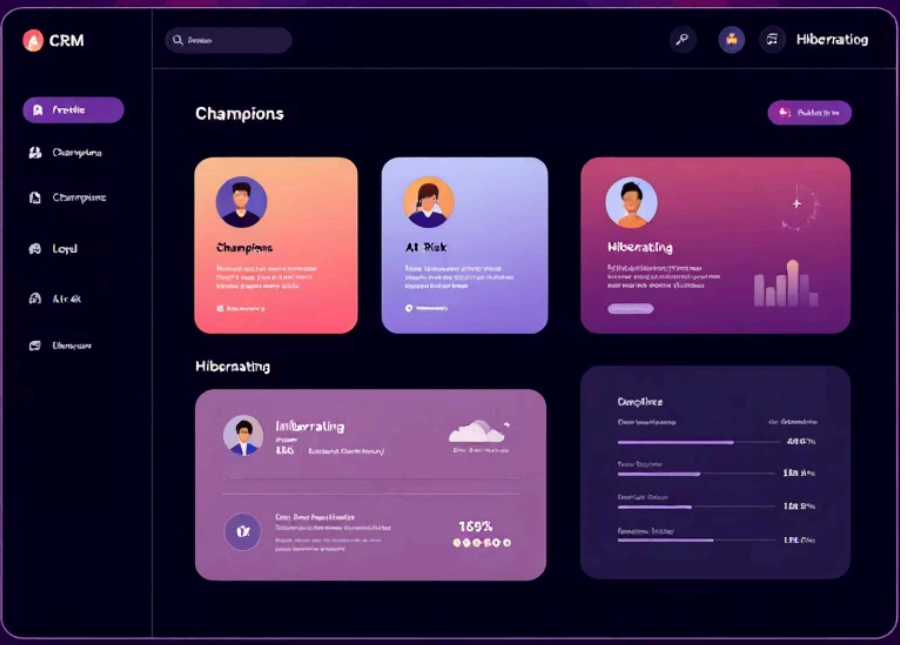
Loyal

Potential Loyal

Medium Risk

High Risk

Made with GAMMA



Model Building

Recommendation Model

Hybrid

Customer Churn Prediction

Made with GAMMA

Deployment Architecture & Deliverables

Scalable Cloud

AWS - (EC2, S3)

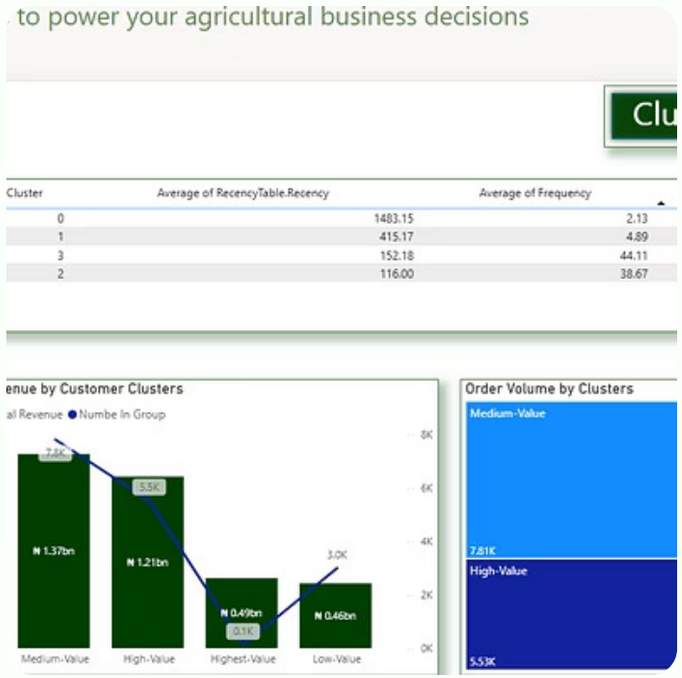
Real-Time Data Flow

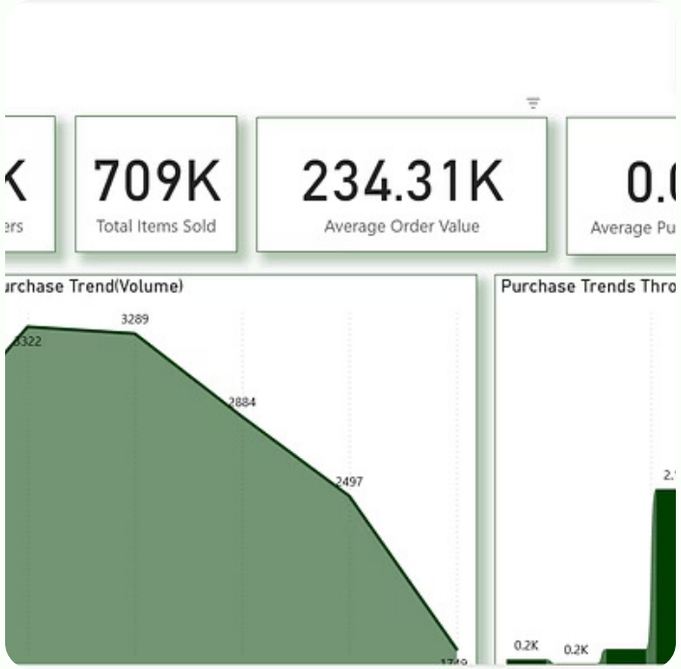
AWS (S3), Amazon glue

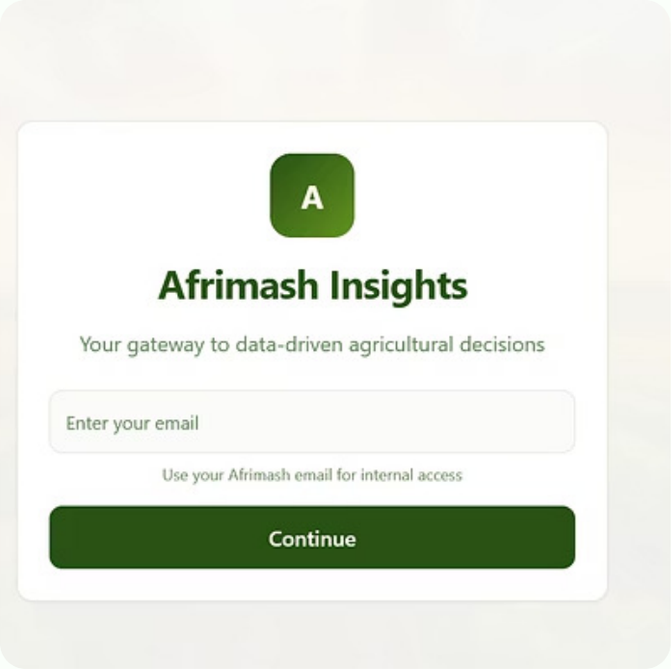
Deliverables

A real-time Power BI dashboard and an interactive web application.

http://13.60.167.68/







Made with GAMMA

Afrimash Customer Intelligence Architecture

End-to-End Data Pipeline & Analytics Platform

Data Sources

RFM Database
PostgreSQL / MySQL
Customer metrics, segments

Transaction DB
PostgreSQL / MongoDB
Order history, products

External APIs
REST / GraphQL
CRM, Marketing data

Data Lake
S3 / Azure Blob
Raw data, backups

ETL Pipeline

Apache Airflow / AWS Glue / Azure Data Factory

Extract | Transform & Clean | Feature Engineering | Validation | Load

Analytics Engine

Customer Segmentation
- RFM Analysis
- K-Means Clustering
- Behavioral Grouping
- Cohort Analysis

Predictive Models
- Churn Prediction
- CLV Forecasting
- Purchase Probability
- Demand Forecasting

Recommendation Engine
- Collaborative Filtering
- Content-Based
- Hybrid Models
- Cross-Sell Analysis

Cloud Platform
AWS / Azure / GCP
Auto-scaling
Load Balancing
VPC / Security

Storage
PostgreSQL 14+
MongoDB
Redis Cache

ML Ops
MLflow / Kubeflow
Model Registry
A/B Testing

Monitoring
Prometheus
Grafana
ELK Stack

Security
TLS 1.3
AES-256
RBAC / IAM

Analytics Engine

Customer Segmentation
- RFM Analysis
- K-Means Clustering
- Behavioral Grouping
- Cohort Analysis

Predictive Models
- Churn Prediction
- CLV Forecasting
- Purchase Probability
- Demand Forecasting

Recommendation Engine
- Collaborative Filtering
- Content-Based
- Hybrid Models
- Cross-Sell Analysis

Monitoring
Prometheus
Grafana
ELK Stack

Security
TLS 1.3
AES-256
RBAC / IAM

API Gateway

FastAPI / Node.js + Express

REST API | GraphQL | Redis Cache | Rate Limiting | OAuth 2.0

CI/CD
GitHub Actions
Docker + K8s
Auto Testing

Presentation Layer

Web Dashboard
React + Recharts + Tailwind

Mobile App
React Native / Flutter

Automated Reports
Email / PDF / Excel

Data Flow: ⟶ Infrastructure: - - - - - -

Made with GAMMA